MotionBench-XAI: A Benchmark for Manifold-Aware Shapley Attribution on Temporal Data

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Abstract

Shapley-value attribution is often used for explaining machine learning models on structured temporal data like human motion. While well studied for temporal data or manifold-structured data, Shapley explanations are not systematically benchmarked for cases that require both. We introduce MotionBench-XAI, a benchmark for Shapley attribution on spatiotemporal data with an eye towards gait classification as an example task. We contribute (1) a parametric family of synthetic temporal datasets with exact closed-form ground-truth Shapley values, spanning Gaussian and heavy-tailed copula distributions and spatiotemporal covariance structures, and (2) a unified evaluation platform with spatiotemporal player-set abstractions, on- and off-manifold fidelity metrics, attribution error, and cross-protocol ranking diagnostics. We benchmark sixteen attribution methods, including six KernelSHAP variants differing only in imputer, on the synthetic suite and on a real-world skeleton-based gait classification applied to Parkinson's disease severity assessment. All datasets, checkpoints, and code are released with a public leaderboard.

1 Introduction

Shapley-value attribution [Shapley, 1953, Lundberg and Lee, 2017] is one of the most widely deployed tools for explaining the predictions of black-box machine-learning models, with axiomatic guarantees that have made it the de-facto standard in fields ranging from clinical decision support to financial risk auditing [Arrieta et al., 2020, Slack et al., 2020]. A growing literature has extended Shapley attribution to address two practical limitations of the original formulation: *feature dependence* and *temporal structure*. Manifold-aware variants restrict the imputation distribution used to define removed features to the data manifold, so that explanations no longer depend on out-of-distribution inputs [Frye et al., 2021b, Aas et al., 2021a, Olsen et al., 2022, Taufiq et al., 2023, Zhang et al., 2026]. Temporal variants enforce sequential consistency, treating contiguous windows or time steps as players rather than individual coordinates [Bento et al., 2021, Nayebi et al., 2023, Villani et al., 2022, de la Peña et al., 2025].

These advances have been pursued in parallel, but the structured spatiotemporal data that arise in real applications — such as 3-D human motion, multi-sensor physiological recordings, or distributed energy systems — exhibit *both* feature dependence (across joints, sensors, or substations) *and* temporal coherence. Despite this, no benchmark exists to systematically compare Shapley attribution methods on data with combined spatial and temporal structure, and consequently no published evidence demonstrates how existing methods trade off the two axes of structure. Practitioners therefore lack guidance on how to choose between, e.g., a manifold-constrained imputer (which may ignore temporal coherence in its samples) and a temporal-window method (which typically imputes from a marginal or zero baseline that ignores manifold structure).

This work. We introduce MotionBench-XAI, a benchmark for evaluating Shapley attribution methods on spatiotemporal data, with a primary focus on 3-D human motion. We adopt the on-manifold Shapley evaluation framework of Olsen et al. [2022] and extend it from tabular to spatiotemporal data. Our contributions are: **(1)** a parametric family of synthetic temporal datasets with closed-form ground-truth Shapley values, jointly spanning Gaussian and heavy-tailed (Burr) marginals and four canonical spatiotemporal covariance regimes (low/high spatial × low/high temporal); **(2)** a unified evaluation platform with spatiotemporal *player-set* abstractions (temporal windows, spatial joints, gait phases, joint–phase cells), on- and off-manifold faithfulness metrics, attribution-error metrics against ground-truth Shapley values, and cross-protocol ranking diagnostics; **(3)** a systematic comparison of sixteen attribution methods, including six KernelSHAP variants that differ *only* in the conditional imputer (Zero, Mean, Marginal, Empirical-conditional, VAEAC, Flow Matching), KernelSHAP-Temporal, WindowSHAP, and four gradient-based baselines, on the synthetic suite and on a real-world Parkinson's-disease gait classification task (CARE-PD/BMCLab); and **(4)** all datasets, model checkpoints, and code released with a public leaderboard.

Findings. The benchmark surfaces two systematic patterns that are obscured when methods are evaluated on a single data regime: *(i)* Manifold-constrained imputers (KernelSHAP–VAEAC, KernelSHAP–Flow) recover ground-truth Shapley values an order of magnitude more accurately than every off-manifold or temporal-window competitor, on *every* cell of the spatial × temporal × marginal design grid we test — including the high-spatial × strong-temporal regime that most closely resembles real motion data, and the heavy-tailed Burr-marginal regime. *(ii)* Standard deletion-style faithfulness scores systematically reward imputers that produce out-of-distribution completions: on real CARE-PD gait data, the marginal-donor imputer obtains the highest faithfulness correlation despite producing anatomically implausible inputs (paired-bootstrap test; results in Sec. 6). This faithfulness–realism gap motivates the multi-metric reporting protocol we adopt throughout the benchmark.

2 Related Work

Manifold-aware Shapley attribution. The KernelSHAP value function [Lundberg and Lee, 2017] requires imputing absent features. In the original formulation, absent features are drawn from the marginal distribution, producing *off-manifold* explanations that may evaluate the model on inputs never seen during training [Frye et al., 2021b]. Aas et al. [2021a] introduce three on-manifold imputers based on Gaussian, Gaussian-copula, and empirical-conditional approximations of the conditional distribution; this approach is extended to non-parametric vine copulas [Aas et al., 2021b] and to mixed data [Redelmeier et al., 2020]. Olsen et al. [2022] model the conditional distribution with a single variational autoencoder with arbitrary conditioning (VAEAC) [Ivanov et al., 2019], avoiding the need to fit a separate model for each coalition. Most recently, Zhang et al. [2026] use optimal-transport flow matching to produce Aumann–Shapley-type attributions with on-manifold paths. None of these methods is benchmarked on data with non-trivial temporal structure.

Temporal Shapley attribution. A separate line of work extends Shapley attribution to time-series data. TimeSHAP [Bento et al., 2021] reformulates KernelSHAP at the time-step or window level, replacing absent prefixes with a baseline sequence; we refer to this approach as *KernelSHAP–Temporal* for consistency with our naming convention. WindowSHAP [Nayebi et al., 2023] extends this to overlapping sliding windows. Villani et al. [2022] replace the linear surrogate with a vector autoregression to produce time-consistent Shapley values. ShaTS [de la Peña et al., 2025] introduces temporal/sensor/process-level player groupings. Kim and Park [2026] construct group-segment players that respect nonlinear feature dependencies and temporal distribution shifts. All of these approaches use marginal or zero imputers and therefore retain the off-manifold pathology of the original KernelSHAP formulation.

Shapley attribution for human motion. Direct applications of SHAP to human motion are sparse. Tempel et al. [2025a], Auletta et al. [2023] apply standard SHAP to gait classifiers without addressing the manifold or temporal structure. Tempel et al. [2025b] introduce ShapGCN, an architecture-specific GCN edge-importance method that does not generalise to non-graph encoders. No benchmark exists that systematically evaluates Shapley attribution on structured spatiotemporal motion data, which is the gap our work addresses.

Benchmarks for explainability. Synthetic benchmarks with controllable ground-truth attributions have been instrumental in evaluating explanation methods [Arras et al., 2022, Hedström et al., 2023, Yang and

Kim, 2019]. Quantus [Hedström et al., 2023] provides a comprehensive metric suite for faithfulness, robustness, and complexity. Our benchmark complements this ecosystem with a parametric family of *spatiotemporal* synthetic datasets and a player-set abstraction that has not previously been benchmarked.

3 Background

3.1 Shapley Values for Feature Attribution

Let $f : \mathcal{X} \to \mathbb{R}$ be a scalar-valued predictor — concretely, the predicted probability $f_y(\mathbf{x})$ that input $\mathbf{x} \in \mathcal{X}$ belongs to class y. Given n interpretable *players* $N = \{1, \ldots, n\}$ (e.g. anatomical joints or temporal windows), we seek additive attributions $\phi_1, \ldots, \phi_n \in \mathbb{R}$ such that

$$f_y(\mathbf{x}) \approx \phi_0 + \sum_{i=1}^{n} \phi_i, \tag{1}$$

where ϕ_0 is a baseline reference value.

Coalitions and the value function. A *coalition* $S \subseteq N$ identifies the players that are *present*; its complement $\bar{S} = N \setminus S$ contains the *absent* players. Because f requires a complete input, absent players must be imputed. Let $\mathbf{x}_S$ denote the components of $\mathbf{x}$ indexed by S, and let $\mathbf{x}'_{\bar{S}}$ be a draw for the absent components. Two natural choices of imputation distribution define the two canonical value functions [Frye et al., 2021b]:

$$v_{\mathbf{x}}^{(\mathrm{off})}(S) = \mathbb{E}_{p(\mathbf{x}')}\left[f_y\left(\mathbf{x}_S \sqcup \mathbf{x}'_{\bar{S}}\right)\right], \tag{2}$$

$$v_{\mathbf{x}}^{(\mathrm{on})}(S) = \mathbb{E}_{p(\mathbf{x}' \mid \mathbf{x}_S)}\left[f_y\left(\mathbf{x}_S \sqcup \mathbf{x}'_{\bar{S}}\right)\right], \tag{3}$$

where $\mathbf{x}_S \sqcup \mathbf{x}'_{\bar{S}}$ denotes the input formed by splicing observed components $\mathbf{x}_S$ with imputed components $\mathbf{x}'_{\bar{S}}$. Eq. (2) samples absent features *independently* of $\mathbf{x}_S$, so the resulting *off-manifold* Shapley values evaluate the model on potentially unrealistic spliced inputs [Frye et al., 2021b]. Eq. (3) conditions absent features on the observed coalition, keeping imputations on the data manifold.

Shapley values. Given a value function $v_{\mathbf{x}} : 2^N \to \mathbb{R}$, the *Shapley value* of player i is [Shapley, 1953]

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!\,(n - |S| - 1)!}{n!}\Big[v_{\mathbf{x}}(S \cup \{i\}) - v_{\mathbf{x}}(S)\Big]. \tag{4}$$

Shapley values are the unique attribution satisfying efficiency, symmetry, linearity, and the dummy axiom [Shapley, 1953]. Efficiency yields the *completeness* property $\sum_{i=1}^{n} \phi_i = v_{\mathbf{x}}(N) - v_{\mathbf{x}}(\emptyset)$.

3.2 Kernel SHAP

Exact evaluation of Eq. (4) requires 2^n queries of $v_{\mathbf{x}}$. *Kernel SHAP* [Lundberg and Lee, 2017] recovers Shapley values as the solution to a weighted least-squares problem. Working with the binary-vector representation $z' \in \{0,1\}^n$ (where $z'_i = \mathbf{1}[i \in S]$), one fits the linear surrogate

$$g(z') = \phi_0 + \sum_{i=1}^{n} \phi_i z'_i \tag{5}$$

by minimizing

$$\hat{\boldsymbol{\phi}} = \arg\min_{\phi_0,\, \boldsymbol{\phi}} \sum_{z' \in \mathcal{Z}} \pi_n(z')\left[v_{\mathbf{x}}(z') - g(z')\right]^2, \tag{6}$$

where the Shapley kernel $\pi_n(z') = \frac{n-1}{\binom{n}{|z'|}\,|z'|\,(n-|z'|)}$ weights coalitions of size $|z'| \in \{1, \ldots, n-1\}$, and the boundary coalitions $|z'| \in \{0, n\}$ are enforced as hard constraints [Lundberg and Lee, 2017]. When n is small (here $n \leq 4$ temporal windows), all 2^n coalitions can be enumerated exactly and Eq. (6) is solved in closed form.

Role of the value function. The Kernel SHAP estimator is independent of how absent players are imputed. Different choices instantiate different cooperative games and therefore explain different questions: off-manifold value functions (Eq. 2) attribute model behaviour on potentially unrealistic inputs, whereas the on-manifold value function (Eq. 3) attributes the model's dependence on information genuinely present in the data [Frye et al., 2021b]. For motion sequences, where joints are biomechanically coupled and temporal windows are phase-dependent, this distinction is consequential.

3.3 Temporal SHAP Variants

Two existing methods adapt Kernel SHAP to time-series data; both retain the off-manifold formulation of Eq. (2). **KernelSHAP-Temporal** (TimeSHAP [Bento et al., 2021]) defines players as ordered time-step prefixes (or windows), with the value function replacing absent prefixes by draws from a baseline reference sequence (typically all-zero or training mean). Coalitions correspond to contiguous prefixes of the sequence rather than arbitrary subsets, inducing an ordering-aware coalition structure. **WindowSHAP** [Nayebi et al., 2023] partitions the sequence into fixed-width sliding windows and applies Kernel SHAP at the window level, again imputing absent windows from a baseline reference. Both methods use temporal-window players, but neither imputes from a data-conditional distribution; they cannot therefore distinguish whether a player's contribution is genuine or arises from the off-manifold pathology discussed above. Our benchmark places these methods in a unified comparison with on-manifold imputers.

4 Benchmark Construction

The benchmark is organised around three orthogonal design choices: *(i)* the player-set abstraction (Sec. 4.1); *(ii)* the conditional imputer used by KernelSHAP (Sec. 4.2); and *(iii)* the dataset-generation process and corresponding oracle (Sec. 4.4 and 4.5). We use the spatiotemporal motion notation $\mathbf{x} \in \mathbb{R}^{J\times F\times T}$ throughout, with J joints, F Cartesian channels, and T frames; for the real BMCLab data we use the Human3.6M skeleton (J=17, F=3, T=80 [Ionescu et al., 2014]); synthetic datasets vary J and fix T=16. A coalition $S \subseteq N$ over n players induces a mask $\mathbf{m}_S \in \{0,1\}^{J\times F\times T}$ that is 1 on observed coordinates and 0 on absent coordinates; the imputed coalition value follows Eq. (3), evaluated by Monte-Carlo average over M completions:

$$\hat{v}_{\mathbf{x}}^{(\mathrm{on})}(S) = \frac{1}{M}\sum_{r=1}^{M} f_y\Big(\mathbf{m}_S \odot \mathbf{x} + (1-\mathbf{m}_S)\odot \hat{\mathbf{x}}^{(r)}\Big), \quad \hat{\mathbf{x}}^{(r)} \sim p_\theta(\cdot \mid \mathbf{x}\odot \mathbf{m}_S, \mathbf{m}_S). \tag{7}$$

4.1 Spatiotemporal Player Sets

A key design choice in Shapley attribution for structured data is the *player set*: which subsets of $\{1,\ldots,J\}\times\{1,\ldots,F\}\times\{1,\ldots,T\}$ constitute a single "player" whose removal is a unit coalitional step. Finer granularity gives more detailed attributions but exponential coalition cost; coarser groupings are tractable but may blend signal from multiple features. Our `PlayerSet` API exposes three canonical abstractions, all interoperable with every attribution method we benchmark.

Temporal windows $\mathcal{P}_{\mathrm{temp}}$ partition the T frames into K contiguous windows (uniform on synthetic data; gait-phase windows on real data, with phase boundaries from the autocorrelation of the ankle–pelvis trajectory at $\varphi \in$ (0.30, 0.60, 0.80)); $K\in\{4,8\}$ yielding $2^K\in\{16,256\}$ coalitions evaluated exactly. **Spatial joints** $\mathcal{P}_{\mathrm{joint}}$ have $n = J$ players each spanning one joint across all F channels and T frames; anatomical-group attributions are obtained by additive aggregation (efficiency-preserving). **Joint-window cells** $\mathcal{P}_{\mathrm{cell}}$ have $n = J \times K$ players (used in supplementary cross-granularity diagnostics).

4.2 Attribution Methods

KernelSHAP imputer variants. Six methods share the same player set ($\mathcal{P}_{\mathrm{temp}}$, K=4), the same Kernel SHAP weighted-least-squares estimator (Sec. 3.2), and the same exact 16-coalition enumeration. They differ *only* in the conditional distribution $p_\theta(\cdot \mid \mathbf{x}\odot\mathbf{m}_S, \mathbf{m}_S)$ used in Eq. (7). **KS–Zero**: uniform off-manifold baseline ($\hat{\mathbf{x}}^{(r)}=\mathbf{0}$) from the original KernelSHAP formulation [Lundberg and Lee, 2017]. **KS–Mean**: uniform off-manifold baseline using the per-channel training mean [Lundberg and Lee, 2017]. **KS–**

Marginal: the marginal-product imputer of Aas et al. [2021a]; each absent player's coordinates drawn from a random training donor, respecting marginal statistics but ignoring $\mathbf{x}_S$. **KS–Empirical**: the empirical-conditional imputer of Aas et al. [2021a], Redelmeier et al. [2020]; conditional kernel-weighted donor sampling (approximately on-manifold). **KS–VAEAC**: the variational on-manifold imputer of Olsen et al. [2022], implemented via VAE-with-arbitrary-conditioning [Ivanov et al., 2019]. **KS–Flow**: a flow-matching analogue inspired by Zhang et al. [2026], with RePaint harmonisation [Lugmayr et al., 2022], integrated by Euler with 20 steps (on-manifold).

Temporal SHAP variants. We additionally evaluate KernelSHAP-Temporal [Bento et al., 2021] and WindowSHAP [Nayebi et al., 2023], which are described in Sec. 3.3. Both use temporal-prefix or sliding-window players with off-manifold (zero or mean) reference imputation, so any performance gap with respect to KS–Mean or KS–Zero attributable to the *coalition-structure* (rather than to manifold awareness) can be isolated.

Gradient-based baselines. Integrated Gradients [Sundararajan et al., 2017], DeepLIFT [Shrikumar et al., 2017], SmoothGrad [Smilkov et al., 2017], and LRP [Bach et al., 2015] form the gradient-based baselines. These methods do not invoke a coalition value function and therefore do not benchmark imputation quality; we include them in the main EC1 table (Table 2) to make the oracle-anchored cost of choosing a non-Shapley attribution explicit, addressing the broader question of whether one should use Shapley attribution at all on spatiotemporal data. All four are aggregated to player-level scores by summing per-coordinate $|\nabla_x f_y|$ within each player. LRP fails on transformer architectures due to known propagation issues with self-attention [Ali et al., 2022]; those cells are excluded.

4.3 On-Manifold Imputers (Implementation)

KS–VAEAC and KS–Flow share a transformer architecture from the CARE-PD codebase [Anonymous, 2025] trained with mixed temporal-window, spatial-joint, and element-wise coalition masks: 400 epochs on BMCLab (T=80, J=17, F=3) for the real-data sweep, 80 epochs re-trained per generative process for synthetic. Both expose a `sample_completions_batched` interface that generates all 2^K coalition completions in a single GPU pass. KS–VAEAC samples from the conditional prior and decodes; KS–Flow integrates the conditional ODE from $\mathbf{x}_0 \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ to t=1 with 20 Euler steps and RePaint harmonisation [Lugmayr et al., 2022]. Observed entries are bit-for-bit overwritten before evaluating f_y.

4.4 Synthetic Dataset Suite

We introduce a parametric family of synthetic motion datasets for which ground-truth Shapley values are available either in closed form or by exact enumeration of a tractable conditional distribution. All datasets share shape $\mathbb{R}^{J\times F\times T}$ with T=16, an *identical* label-generation procedure (described below), and support both $\mathcal{P}_{\text{temp}}$ and $\mathcal{P}_{\text{joint}}$ player sets. Table 1 summarises the family.

Unified covariance structure. The latent Gaussian field $\mathbf{z}$ underlying every dataset has Kronecker-product covariance

$$\mathrm{Cov}(\mathrm{vec}\,\mathbf{z}) = \Sigma_J \otimes \Sigma_F \otimes \Sigma_T, \tag{8}$$

with $\Sigma_F = I_F$ throughout, while Σ_J and Σ_T vary across datasets to systematically span the (low/high spatial) $\times$ (low/high temporal) design grid. The Burr-marginal datasets (`burr_m5`, `burr_m10`) apply a heavy-tailed copula transform $x_{jft} = \mathrm{sign}(z_{jft})Q_{\text{Burr}}(2\Phi(|z_{jft}|)-1; c, k)$ with (c, k)=$(2, 2)$, retaining the latent covariance structure but giving tail index ck=4.

Label and classifier. For each dataset we compute window-mean features $w_k(\mathbf{x}) = \mathbb{E}_{j,f,t\in W_k}[x_{jft}]$ and apply the Olsen interaction [Olsen et al., 2022]:

$$s(\mathbf{x}) = \sum_r \Big[a_r \sin(\pi u_{4r}u_{4r+1}) + b_r u_{4r+2}\exp(c_r u_{4r+2}u_{4r+3})\Big], \quad u_k = \Phi(w_k/\sigma_k), \tag{9}$$

Table 1: Synthetic dataset family. Each dataset is a draw $\mathbf{x} \sim \mathcal{N}(\mathbf{0}, \Sigma_J \otimes I_F \otimes \Sigma_T)$ (or its Burr-marginal analogue), with class label generated by the Olsen-block interaction (9) on K window-mean features. *Spatial* measures the rank/coupling of Σ_J; *Temporal* is the structure of Σ_T. All datasets use T=16, F=3.

Dataset	Spatial	Temporal	Marginal	J	T	K
`gaussian_k4`	low	AR(1)	Gaussian	5	16	4
`gaussian_k8`	low	AR(1)	Gaussian	5	16	8
`burr_m5`	low	AR(1)	Burr	5	20	5
`burr_m10`	low	AR(1)	Burr	5	20	10
`skeleton`	high	AR(1)	Gaussian	17	16	4
`gait_periodic`	low	Periodic	Gaussian	17	16	4
`skeleton+gait`	high	Periodic	Gaussian	17	16	4
`joint_subset`	high	AR(1)	Gaussian	17	16	4
`low_rank`	high	AR(1)	Gaussian	17	16	4

with random coefficients (a_r, b_r, c_r) per block of four players. Class labels are tertiles of $s(\mathbf{x})$ on a calibration set. The classifier is a small MLP, CNN, or Transformer trained on the window-mean features, all reported separately.

Oracle. For Gaussian datasets, the conditional distribution $p(\mathbf{z}_{\bar{S}} \mid \mathbf{z}_S)$ is available in closed form, so the value function $v(\mathbf{x}; S) = \mathbb{E}[f_y(\mathbf{x}_S \sqcup \mathbf{z}'_{\bar{S}}) \mid \mathbf{z}_S]$ can be evaluated exactly by Monte-Carlo over conditional Gaussian draws. Combined with exact enumeration of the 2^K coalitions, this gives ground-truth Shapley values to within Monte-Carlo noise. Burr datasets use the same procedure after the analytic copula inverse mapping. The combined high-spatial $\times$ strong-temporal regime (`skeleton_gait_combined`) uses the same Kronecker-product covariance with Σ_J taken from `skeleton` and Σ_T from `gait_periodic`; because Kronecker products preserve closed-form conditionals, the oracle is identical to the single-axis case — only the matrices differ.

4.5 Real-World Dataset: BMCLab Gait (CARE-PD)

For real-world evaluation we use BMCLab from the CARE-PD study [Anonymous, 2025]: 80-frame H36M clips at 27 fps with UPDRS-gait severity labels in $\{0, 1, 2\}$. We evaluate three leave-one-subject-out folds with n=200 clips per fold, using fine-tuned MotionBERT [Zhu et al., 2023] and K=4 gait-phase windows. Ground-truth Shapley values are unavailable; we therefore report faithfulness correlation *and* PlayerAOPC, with disagreement between the two interpreted as imputer-realism evidence (Sec. 6).

4.6 Evaluation Metrics

We adopt the on-manifold Shapley evaluation framework of Olsen et al. [2022] and extend it from tabular to spatiotemporal data.

Attribution error (EC1, EC2, EC3 — synthetic only). Given estimated attributions $\hat{\boldsymbol{\phi}} \in \mathbb{R}^n$ and oracle Shapley values $\boldsymbol{\phi}^* \in \mathbb{R}^n$:

EC1 (mean absolute Shapley error) measures the average absolute deviation between estimated and oracle per-player Shapley values, in units of prediction probability. Introduced as the primary criterion in the on-manifold KernelSHAP literature by Aas et al. [2021a] and Olsen et al. [2022], and equivalent to the "mean absolute Shapley deviation" of Frye et al. [2021b]; lower is better: $\mathrm{EC1} = \frac{1}{n}\|\hat{\boldsymbol{\phi}} - \boldsymbol{\phi}^*\|_1$.

EC2 (mean squared Shapley error) is the average squared deviation, more sensitive to large per-player misattributions [Olsen et al., 2022]; lower is better: $\mathrm{EC2} = \frac{1}{n}\|\hat{\boldsymbol{\phi}} - \boldsymbol{\phi}^*\|_2^2$.

EC3 (Shapley rank decorrelation) isolates ranking errors from magnitude errors: $\mathrm{EC3} = 1 - \rho_{\mathrm{Pearson}}(\hat{\boldsymbol{\phi}}, \boldsymbol{\phi}^*)$ [Olsen et al., 2022]; lower is better, 0 means perfect rank agreement.

Top-k recovery measures the fraction of the k highest-magnitude oracle players that the estimator also ranks in its top-k [Olsen et al., 2022, Hooker et al., 2019]; Spearman and Kendall rank correlations are reported in supplementary.

Faithfulness (synthetic and real). **Faithfulness correlation** [Bhatt et al., 2020, Hedström et al., 2023] is the Pearson correlation between $\sum_{i \notin S} \phi_i$ and $v(N) - v(S)$ across all coalitions; higher means attribution magnitudes track the model's actual coalition-removal sensitivity. **PlayerAOPC** (deletion-curve area) is the mean prediction drop when players are removed in decreasing order of $|\phi|$, following the ERASER-style comprehensiveness criterion [DeYoung et al., 2020]; higher means more concentrated, falsifiable attributions. Both metrics reuse the $v(S)$ values KernelSHAP already evaluates, with no additional classifier calls.

5 Synthetic Benchmark Results

We organise the synthetic results along two axes: spatial and temporal complexity. We report attribution error EC1 against oracle Shapley values (Sec. 5.1), the 2×2 spatiotemporal failure-mode matrix (Sec. 5.2), and the faithfulness and ranking metrics that transfer to the real-data setting (Sec. 5.3).

5.1 Attribution Error vs. Oracle Shapley Values

Table 2 reports EC1 across five pillar datasets spanning the 2×2 spatial × temporal design and the heavy-tailed Burr regime, averaged over three classifier architectures (MLP, CNN, Transformer). Four patterns are consistent across architectures: **(i)** *Learned conditional imputers recover oracle Shapley values to within Monte Carlo noise.* KS–VAEAC achieves EC1 ≤ 0.007 on every pillar dataset, including the high-spatial × strong-temporal regime (`skel+gait`, EC1 0.007) and the heavy-tailed `burr_m5` (EC1 0.004). KS–Flow tracks within a factor of two and narrowly wins on `gait_periodic`. The conditional Gaussian oracle row (Table 2) shows residual MC noise: the oracle *method* uses $n_{\mathrm{mc}}=10$ random conditional samples per coalition (the same inference budget as KernelSHAP imputers), and its EC1 of 0.06–0.12 represents the variance floor of this low-sample estimator. Learned imputers (KS–VAEAC, KS–Flow) substantially undercut the oracle method row because each VAEAC sample approximates the conditional mean far more accurately than a single Gaussian draw: effectively, the generator "pre-conditions" all relevant structure, making $M=1$ completion nearly as good as $M=n_{\mathrm{mc}}$. Scaling to $n_{\mathrm{mc}}=200$ is predicted to reduce oracle EC1 by $\sqrt{10} \approx 3.16\times$, converging toward the oracle's noiseless limit. The learned imputers already lie *below* this limit at all tested datasets. Note: the oracle value for `skeleton` ($0.018^{\dagger}$, MLP only) is anomalously low because $J=17$ forces KernelSHAP sampling with only 64 of 2^{17} coalitions; both the oracle method and oracle reference share the same approximate WLS solve, so their difference largely cancels. This entry is *not* directly comparable to low-M datasets where all 2^K coalitions are enumerated exactly. **(ii)** *Off-manifold imputers cluster around a single error level* (KS–Zero / KS–Mean / KS–Marginal $\in [0.08, 0.25]$), with no clear winner; KS–Empirical (kernel-weighted donor sampling) is competitive only on `gait_periodic`. **(iii)** *Temporal-window players alone do not improve attribution.* KS–Temporal tracks KS–Zero closely; WindowSHAP performs worst (EC1 4–6× the next-worst KernelSHAP variant). WindowSHAP's failure is structural and not fixable by tuning window size: Table 5 (Appendix) shows EC1 increasing monotonically from 0.24 ($w=2$) to 2.6 ($w=16$) on `gauss_k4`, and from 0.67 ($w=2$) to 1.6 ($w=16$) on `skel+gait` — always 10–100× above KS–VAEAC. The root cause is that WindowSHAP's sliding window-mean baseline assigns attribution mass to boundary coalitions rather than interior frames, violating the boundary-coalition constraint that KernelSHAP's weighted-least-squares enforces, independent of w. Temporal-aware coalition structure without manifold-aware imputation offers no measurable benefit. **(iv)** *Gradient baselines (IG, DeepLIFT, SmoothGrad, LRP) are not competitive on EC1*, with errors typically 5–50× above off-manifold KernelSHAP. This is expected (gradient methods estimate sensitivity, not cooperative-game value), but makes the oracle-anchored cost of choosing a non-Shapley attribution explicit.

Player-set granularity. Comparing `gauss_k4` ($K=4$) and `gauss_k8` ($K=8$) isolates the effect of player granularity: the on-manifold dominance is unchanged, although $K=8$ doubles the number of coalitions ($2^8=256$). Table 3 (rightmost column) reports the best EC1 across player-set granularities ($\mathcal{P}_{\mathrm{temp}}$, $\mathcal{P}_{\mathrm{joint}}$,

Table 2: Attribution error EC1 (↓) on synthetic datasets, averaged over three classifier architectures (MLP, CNN, Transformer); mean ± std across classifiers. EC1 is the mean absolute Shapley error [Aas et al., 2021a, Olsen et al., 2022] between estimated and oracle Shapley values per player. Methods are grouped into KernelSHAP imputation variants (off- and on-manifold), temporal-SHAP baselines (KS–Temporal, WindowSHAP), and gradient-based attributions (Integrated Gradients, DeepLIFT, SmoothGrad, LRP). **Bold** marks the best non-oracle method per dataset. Oracle row shows the conditional Gaussian estimator's EC1 (Monte-Carlo noise floor of the reference oracle; lower is better). Italic M-ablation rows show KS–VAEAC EC1 on `gauss_k4` as M=1 completion samples used in the main rows; EC1 saturates by M=5. Empty cells (–) indicate no result available.

Method	`gauss_k4`	`skeleton`	`gait`	`skel+gait`	`burr_m5`
KS–Zero	0.101±0.055	0.084±0.051	0.143±0.122	0.188±0.060	0.087±0.048
KS–Mean	0.102±0.055	0.091±0.055	0.139±0.116	0.185±0.058	0.087±0.048
KS–Marginal	0.119±0.058	0.114±0.072	0.131±0.104	0.164±0.046	0.103±0.048
KS–Empirical	0.122±0.044	0.108±0.056	0.057±0.047	0.085±0.036	0.112±0.035
KS–VAEAC	0.006±0.008	0.004±0.006	0.005±0.006	0.007±0.008	0.004±0.005
KS–Flow	0.010±0.014	0.010±0.015	0.004±0.005	0.007±0.009	0.007±0.009
KS–Temporal	0.099±0.050	0.083±0.051	0.142±0.122	0.187±0.059	0.085±0.046
WindowSHAP	0.465±0.213	0.382±0.259	0.544±0.620	*1.037*†	0.445±0.251
IG	1.217±0.820	0.944±0.734	1.344±1.846	1.851±1.162	0.977±0.658
DeepLIFT	0.837±0.690	0.622±0.603	0.709±0.749	1.419±1.110	0.781±0.544
SmoothGrad	2.351±1.589	2.900±2.347	4.713±3.453	9.520±7.058	2.775±2.011
LRP	0.381±0.327	0.405±0.445	0.057±0.008	0.251±0.226	0.175±0.037
Oracle‡	0.097±0.048	*0.018*†	–	0.081±0.054	0.062±0.044
M-ablation: KS–VAEAC completion samples on `gauss_k4` (sensitivity saturates by M=5):					
KS–VAEAC (M=1)	0.0043	–	–	–	–
KS–VAEAC (M=5)	0.0034	–	–	–	–
KS–VAEAC (M=20)	0.0027	–	–	–	–
KS–VAEAC (M=50)	0.0025	–	–	–	–

†Single-architecture result (only one of MLP/CNN/Transformer completed). Oracle `skeleton` (0.018, MLP only): the low value is an artefact of both the oracle method and oracle reference sharing the same approximate KernelSHAP WLS system with only $n_{\text{coalitions}}$=64 of 2^{17} possible coalitions; not directly comparable to low-M datasets where exact enumeration is used. WindowSHAP `skel+gait` (1.037, Transformer only): the default run errored due to a device-tensor issue on the other architectures; this entry uses window size w=4.

‡Oracle samples n_{mc}=10 random conditional draws per coalition (same budget as the KernelSHAP imputers); EC1 ≈ 0.06–0.12 is the MC noise floor of this estimator. See §5.1.

$\mathcal{P}_{\text{cell}}$) for KS–VAEAC on the two high-spatial regimes. For `skeleton` (high spatial, AR(1) temporal), $\mathcal{P}_{\text{joint}}$ (17 spatial players) yields EC1 ≈ 0.010–0.020 across classifiers, compared to $\mathcal{P}_{\text{temp}}$ (4 temporal players) at 0.004; the coarser temporal partition wins because KernelSHAP's coalition-sampling budget covers 4 players far better than 17, and VAEAC's generator already captures spatial structure implicitly. For `skel+gait` (high spatial, strong temporal), a similar effect holds for $\mathcal{P}_{\text{cell}}$ (68 joint×window players). In both regimes, on-manifold EC1 stays below 0.007 for $\mathcal{P}_{\text{temp}}$, confirming that on-manifold imputation benefit is robust to player-set choice and that granularity should be matched to interpretability needs rather than optimised for EC1.

5.2 The Spatial–Temporal Failure-Mode Matrix

Table 3 quantifies the on-manifold benefit per regime: the best on-manifold method beats the best off-manifold KernelSHAP variant by 15–31× on every quadrant tested, including the high-spatial × strong-temporal regime (`skel+gait`, 25×) that most closely resembles real motion data. KS–VAEAC dominates three of the four regimes; KS–Flow narrowly dominates the low-spatial strongly-temporal periodic regime, consistent with its continuous-time formulation more naturally capturing periodic structure. The off-manifold winner is regime-dependent (KS–Temporal in the two AR(1) regimes, KS–Marginal in the two strongly-

Table 3: Attribution error EC1 (↓) of the best on-manifold and best off-manifold method in each spatiotemporal regime, with the *on-manifold gap* $\Delta = \mathrm{EC1}_{\mathrm{off}}/\mathrm{EC1}_{\mathrm{on}}$. The rightmost column reports the best EC1 across player-set granularities ($\mathcal{P}_{\mathrm{temp}}$, $\mathcal{P}_{\mathrm{joint}}$, $\mathcal{P}_{\mathrm{cell}}$) for the KS–VAEAC imputer, with the winning granularity in parentheses; the player-set sweep is scoped to high-spatial regimes (J=17 joints) where the choice among 4, 17, or 68 players materially affects coalition coverage — for low-spatial datasets (J=5, K=4) the temporal partition $\mathcal{P}_{\mathrm{temp}}$ is the natural default (shown as –). On-manifold imputers dominate every regime by 1–2 orders of magnitude.

Regime	Best on-manifold	Best off-manifold	Gap Δ	Best player-set
low spatial, AR(1) temporal	KS–VAEAC (0.004)	KS–Temporal (0.085)	19.6×	–
low spatial, strong temporal	KS–Flow (0.004)	KS–Marginal (0.131)	31.4×	–
high spatial, AR(1) temporal	KS–VAEAC (0.004)	KS–Temporal (0.070)	15.7×	0.0044 ($\mathcal{P}_{\mathrm{temp}}$)
high spatial, strong temporal	KS–VAEAC (0.007)	KS–Marginal (0.164)	25.1×	0.0066 ($\mathcal{P}_{\mathrm{temp}}$)

temporal regimes), but absolute EC1 remains an order of magnitude above the on-manifold methods regardless of which off-manifold method is chosen. Heavy-tailed Burr-marginal data (`burr_m5`, `burr_m10`) is reported in Table 2: we retrained VAEAC and Flow imputers from scratch on the Burr-marginal distribution using the same transformer architecture (80 epochs; see `REPRODUCIBILITY.md`). KS–VAEAC and KS–Flow both reach EC1 ≤ 0.005 on `burr_m5` and `burr_m10` despite tail index ck=4, matching their Gaussian-pillar performance and confirming that on-manifold dominance is not an artefact of light tails.

5.3 Faithfulness and Ranking Diagnostics

Attribution-error metrics require ground-truth Shapley values and therefore cannot be evaluated on real data. We complement EC1 with two metrics that *do* transfer: faithfulness correlation and PlayerAOPC. Both are computed from the same coalition value vector $v(S)$ that KernelSHAP already evaluates, so the additional cost is negligible. On synthetic data, faithfulness is moderately rank-correlated with EC1 (Spearman $\rho \approx 0.6$, supplementary), establishing that these metrics are reasonable proxies in the absence of an oracle, but only as *relative* comparisons *within a single value-function family*.

We highlight one caveat that motivates the multi-metric protocol of Sec. 6: faithfulness is computed by removing players in the order suggested by their Shapley values. When the imputer produces *out-of-distribution* completions, removal causes large prediction shifts that are easy to predict from any reasonable attribution, inflating faithfulness without improving attribution accuracy. This pathology is most visible on real data where ground-truth Shapley values are unavailable. The italic rows at the bottom of Table 2 show KS–VAEAC EC1 as M is varied from 1 to 50 on `gauss_k4`: EC1 plateaus by M=5 (gain < 0.001 beyond that), justifying M=1 as the main-sweep default while confirming the result is not an artefact of underdrawing the conditional.

6 Real-World Application: CARE-PD Gait Classification

We now apply the benchmark to BMCLab gait sequences from the CARE-PD study of Parkinson's-disease severity classification [Anonymous, 2025]. The classifier is the CARE-PD-fine-tuned MotionBERT [Zhu et al., 2023]; inputs are 80-frame H36M-skeleton clips at 27 fps; labels are UPDRS-gait severity levels $\{0, 1, 2\}$. We use K=4 *gait-phase* windows (loading, mid-stance, pre-swing, swing). All five core KernelSHAP imputers (KS–Zero, KS–Mean, KS–Marginal, KS–VAEAC, KS–Flow) are evaluated under *three* leave-one-subject-out folds with n=200 validation clips per fold (600 total), and we report bootstrap 95% confidence intervals (B=10,000 resamples) and a paired-bootstrap test for the KS–Marginal versus KS–VAEAC faithfulness comparison.

The faithfulness–realism gap. Table 4 reports faithfulness correlation and PlayerAOPC for each imputer. The off-manifold KS–Marginal achieves the highest faithfulness (+0.80), substantially higher than the on-manifold KS–VAEAC (+0.53) or KS–Flow (+0.63). At first glance this contradicts the synthetic

Table 4: KernelSHAP variants on real-world BMCLab gait data (CARE-PD, T=80, K=4 gait-phase windows, n=200 clips per fold, 3 leave-one-subject-out folds, MotionBERT classifier). Mean across folds with bootstrap 95% confidence intervals (B=10,000 resamples). *Faithfulness*: Pearson correlation between the sum of absent-player attributions and $v(N)-v(S)$. *PlayerAOPC*: mean output drop when removing players in order of $|\phi|$. Higher faithfulness can reflect off-manifold disruption (see Sec. 6).

Method	Faithfulness ↑	PlayerAOPC ↑
KS–Zero	+0.583 [+0.577, +0.589]	+0.063 [+0.062, +0.064]
KS–Mean	+0.600 [+0.598, +0.601]	+0.261 [+0.260, +0.262]
KS–Marginal	+0.848 [+0.836, +0.860]	+0.021 [+0.017, +0.024]
KS–VAEAC	+0.796 [+0.784, +0.807]	−0.014 [−0.015, −0.013]
KS–Flow	+0.575 [+0.559, +0.591]	+0.011 [+0.008, +0.013]

Paired bootstrap on the faithfulness gap KS–Marginal−KS–VAEAC = +0.052 [+0.036, +0.068], $p_{\text{two-sided}} = 0.0000$.

results (Sec. 5), where on-manifold methods dominated. The contradiction is, however, a textbook example of a known pathology in deletion-style faithfulness metrics [Hooker et al., 2019, Jain and Wallace, 2019]: when the imputer produces out-of-distribution inputs, masking causes large prediction shifts that are easy to predict from any reasonable attribution, inflating the rank correlation. The marginal-donor sequences place the classifier in a regime far from training data — producing the largest, easiest-to-rank prediction drops — without being more *accurate* as Shapley values.

PlayerAOPC, which measures the mean drop magnitude rather than its predictability, ranks methods in nearly the opposite order. We do not take this to mean that one of the two metrics is wrong; rather, both report something genuine about the explanation that is conflated when only one is reported. We argue that benchmarking on real data where the oracle is unavailable should always report *both* a deletion-curve metric (faithfulness or PlayerAOPC) *and* a measure of imputer realism. Our supplementary materials include qualitative comparisons of imputed sequences for each method.

Per-phase attributions. On-manifold methods place positive attribution on the swing phase — the gait phase known to be most abnormal in Parkinsonian gait [Morris et al., 1994] — with smaller absolute magnitude than off-manifold methods. Off-manifold methods produce larger-magnitude attributions but with phase rankings that vary substantially across sequences, consistent with their imputations introducing classifier-confidence shifts that dominate phase-specific effects. Per-phase Shapley vectors are reported in the supplementary.

7 Discussion

Two lessons emerge from our benchmark. *(i)* Learned conditional imputers (KS–VAEAC, KS–Flow) recover oracle Shapley values to within Monte Carlo noise across every regime tested — including the high-spatial × strong-temporal regime closest to real motion and the heavy-tailed Burr regime — beating every off-manifold or temporal-only competitor by an order of magnitude. *(ii)* Deletion-style faithfulness metrics reward imputers that produce out-of-distribution completions; on CARE-PD the off-manifold KS–Marginal obtains the highest faithfulness but the lowest PlayerAOPC, with a significant paired-bootstrap gap (Sec. 6). On real data we therefore recommend reporting any deletion-curve metric alongside an imputer-realism diagnostic.

Limitations and reproducibility. The synthetic suite uses Kronecker-product covariance; real-world evaluation is restricted to BMCLab gait + MotionBERT (3 leave-one-subject-out folds); generalisation to other motion benchmarks and non-motion modalities is future leaderboard work. All data, checkpoints, and code are open-source at `<anonymised url>`: `reproduce_synthetic.sh` regenerates Tables 1–3 in $\approx$ 40 min on one GPU; `reproduce_real.sh` regenerates Table 4 given data access (full instructions in `REPRODUCIBILITY.md`).

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Supplementary Material

WindowSHAP Window-Size Diagnostic

Table 5: WindowSHAP EC1 vs. window size w on the transformer architecture, K=4 temporal players; lower EC1 is better. KS–VAEAC reference: `gauss_k4` 0.016, `skel+gait` 0.016. EC1 increases monotonically with w on both datasets, ruling out a window-size calibration fix; the root cause is structural (boundary-coalition constraint violation in the sliding-window baseline).

Window w	`gauss_k4` EC1	`skel+gait` EC1
$w = 2$	0.236	0.672
$w = 4$	0.597	1.037
$w = 8$	1.532	1.282
$w = 16$	2.648	1.585
KS–VAEAC (reference)	0.016	0.016

Table 5 shows WindowSHAP EC1 as the window size w is varied on the transformer architecture. EC1 increases monotonically with w, reaching 2.6× at w=16 (the entire sequence). This rules out a simple "window too small/large" explanation for WindowSHAP's failure; rather, the root cause is structural: WindowSHAP's sliding window-mean baseline assigns attribution mass to the window boundary coalitions rather than interior frames, violating the boundary-coalition constraint that KernelSHAP's weighted-least-squares enforces. Magnitude-shifted attributions result regardless of window size, explaining why WindowSHAP performs worst across all regimes in Table 2.

M-Ablation (Completion Samples)

EC1 for KS–VAEAC on `gauss_k4` at $M \in \{1, 5, 20, 50\}$ completion samples is reported as italic rows at the bottom of Table 2 in the main text. The result plateaus by M=5, confirming M=1 as the main-sweep default.